

# MAT1575, Classwork5, Fall2026

## 1.5 Substitution (p.74-81)

1. Evaluate each of the following indefinite integrals:

(a)  $\int (x^2 - 3)^2 dx$

(b)  $\int (x^2 - 3)^3 2x dx$  ?

$$\int (x^4 - 6x^2 + 9) dx = \frac{x^5}{5} - 2x^3 + 9x + C$$

Observation: Let  $g(x) = x^2 - 3$ . Then  $(x^2 - 3)^3 \cdot 2x = [g(x)]^3 \cdot g'(x)$ . Can it be used to get the results in an easier way?

2. Theorem: **u-Substitution** or **Change of Variables** with Indefinite Integrals.

Let  $u = g(x)$ , where  $g'(x)$  is continuous over an interval, let  $f(x)$  be continuous over the corresponding range of  $g$ , and let  $F(x)$  be an antiderivative of  $f(x)$ . Then

$$\int f[g(x)]g'(x)dx = \int f(u)du = F(u) + c = F(g(x)) + c.$$

Proof: Since  $\frac{d}{dx} F(g(x)) = F'(g(x)) \cdot g'(x) = f(g(x))g'(x)$ , then  $\int f(g(x))g'(x)dx = F(g(x)) + C$

Let  $u = g(x)$ . Then  $du = g'(x) dx$  and  $\int f(u)du = F(u) + C$

Note: what does  $du = g'(x)dx$  mean?

$$\Delta x = x_i - x_{i-1} \text{ and } \Delta u = g(x_i) - g(x_{i-1}). \quad \frac{\Delta u}{\Delta x} = \frac{g(x_i) - g(x_{i-1})}{x_i - x_{i-1}} \Rightarrow \Delta u = \frac{g(x_i) - g(x_{i-1})}{x_i - x_{i-1}} \Delta x \Rightarrow du = g'(x) dx$$

3. Using u-Substitution to Find an Antiderivative of the given indefinite integrals:

(a)  $\int (x^2 - 3)^3 2x dx$

$u = x^2 - 3$   
 $du = 2x \cdot dx$

$$\int u^3 du = \frac{u^4}{4} + C = \frac{(x^2 - 3)^4}{4} + C$$

(b)  $\int 6y(3y^2 + 4)^4 dy$

$u = 3y^2 + 4$   
 $du = 6y dy$

$$\int u^4 du = \frac{u^5}{5} + C = \frac{(3y^2 + 4)^5}{5} + C$$

(c)  $\int z\sqrt{z^2 - 5} dz$

$u = z^2 - 5$   
 $du = 2z dz$   
 $\frac{du}{2} = z dz$

$$\begin{aligned} &= \int \sqrt{u} \frac{du}{2} \\ &= \frac{1}{2} \int u^{\frac{1}{2}} du \\ &= \frac{1}{2} \left( \frac{2}{3} u^{\frac{3}{2}} \right) + C = \frac{1}{3} (\sqrt{z^2 - 5})^3 + C \end{aligned}$$

4. Problem-Solving Strategy of **Integration by Substitution**.

- (1) Choose  $u = g(x)$  carefully such that  $du = g'(x)dx$  is a part of integration
- (2) Substitute  $u = g(x)$  and put  $du = g'(x)dx$  into the integration
- (3) Find the antiderivative in terms of  $u$
- (4) Write the results in terms of  $x$  by replacing  $u$  by  $g(x)$

5. Using u-Substitution to Find an Antiderivative of the given indefinite integrals:

(a)  $\int \frac{\sin t}{\cos^3 t} dt$

$u = \cos(t)$   
 $du = -\sin(t) dt$   
 $-du = \sin(t) dt$

$$\begin{aligned} &= \int \frac{-du}{u^3} = -\int \frac{du}{u^3} = -\int u^{-3} du \\ &= -\left( \frac{u^{-2}}{-2} \right) + C = \frac{u^{-2}}{2} + C = \frac{1}{2u^2} + C \\ &= \frac{1}{2 \cos^2(t)} + C \end{aligned}$$

(b)  $\int \cos^3 \theta \sin \theta d\theta$

$u = \cos \theta$   
 $du = -\sin \theta d\theta$   
 $-du = \sin \theta d\theta$

$$\begin{aligned} &= \int u^3 \cdot (-du) = -\int u^3 du \\ &= -\frac{u^4}{4} + C = -\frac{(\cos \theta)^4}{4} + C \end{aligned}$$

(c)  $\int \frac{x}{\sqrt{x-1}} dx$

$u = x - 1$   
 $du = dx$

$$\begin{aligned} &\frac{u}{\sqrt{u}} = \frac{(\sqrt{u})^2}{\sqrt{u}} = \frac{\sqrt{u} \cdot \sqrt{u}}{\sqrt{u}} = \sqrt{u} \\ &\int \frac{u+1}{\sqrt{u}} du = \int \left( \frac{u}{\sqrt{u}} + \frac{1}{\sqrt{u}} \right) du \\ &= \int u^{\frac{1}{2}} + u^{-\frac{1}{2}} du \\ &= \frac{2}{3} u^{\frac{3}{2}} + 2u^{\frac{1}{2}} + C \\ &= \frac{2}{3} (\sqrt{x-1})^3 + 2(\sqrt{x-1}) + C \end{aligned}$$

6. Theorem: **u-Substitution or Change of Variables** with Definite Integrals.

Let  $u = g(x)$ , where  $g'(x)$  is continuous over an interval, let  $f(x)$  be continuous over the corresponding range of  $g$ , and let  $F(x)$  be an antiderivative of  $f(x)$ . Then

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Observation: If we change variables in the integrand, the limits of integration changes as well

7. Using u-Substitution to Find an Antiderivative of the given definite integrals:

(a)  $\int_0^1 x^2(1+2x^3)^5 dx$

$u(x) = 1+2x^3, du = 6x^2 dx$   
 $u(1) = 3$   
 $u(0) = 1$   
 $\frac{du}{6} = x^2 dx$

$$\int_{u(0)}^{u(1)} u^5 \cdot \frac{du}{6} = \int_1^3 \frac{u^5}{6} du$$

$$= \frac{1}{6} \int_1^3 u^5 du = \frac{1}{6} \left( \frac{u^6}{6} \right)_1^3$$

$$= \frac{1}{6} \left[ \frac{3^6}{6} - \frac{1^6}{6} \right] = \frac{728}{36} = \frac{182}{9}$$

(b)  $\int_{-1}^0 y(2y^2-3)^5 dy = \frac{91}{3}$

$u = 2y^2-3, du = 4y dy$   
 $u(0) = -3$   
 $u(-1) = -1$   
 $\frac{du}{4} = y dy$

$$\int_{u(-1)}^{u(0)} u^5 \cdot \frac{du}{4} = \int_{-1}^{-3} \frac{u^5}{4} du$$

$$= \frac{1}{4} \left[ \frac{u^6}{6} \right]_{-1}^{-3}$$

$$= \frac{1}{4} \left[ \frac{(-3)^6}{6} - \frac{(-1)^6}{6} \right] = \frac{728}{24}$$

(c)  $\int_0^1 x e^{4x^2+3} dx$

$u = 4x^2+3, du = 8x dx$   
 $u(1) = 7$   
 $u(0) = 3$   
 $\frac{du}{8} = x dx$

$$= \int_3^7 e^u \frac{du}{8}$$

$$= \frac{1}{8} e^u \Big|_3^7$$

$$= \frac{1}{8} (e^7 - e^3)$$

8. Using u-Substitution to Find an Antiderivative of the given definite integrals:

(a)  $\int_0^1 x^2 \cos\left(\frac{\pi}{2}x^3\right) dx$

$u = \frac{\pi}{2}x^3, du = \frac{3\pi}{2}x^2 dx$   
 $u(1) = \frac{\pi}{2}$   
 $u(0) = 0$   
 $\frac{2}{3\pi} du = x^2 dx$

$$= \int_0^{\frac{\pi}{2}} \cos(u) \cdot \frac{2}{3\pi} du$$

$$= \frac{2}{3\pi} \int_0^{\frac{\pi}{2}} \cos(u) du$$

$$= \frac{2}{3\pi} \left[ \sin(u) \right]_0^{\frac{\pi}{2}}$$

$$= \frac{2}{3\pi} \left[ \sin\left(\frac{\pi}{2}\right) - \sin(0) \right]$$

$$= \frac{2}{3\pi} [1 - 0] = \frac{2}{3\pi}$$

(b)  $\int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta$

$\cos^2 \theta = \frac{1+\cos(2\theta)}{2}$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} + \frac{\cos(2\theta)}{2} d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} d\theta + \int_0^{\frac{\pi}{2}} \frac{\cos(2\theta)}{2} d\theta$$

$$= \frac{\theta}{2} \Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \frac{\cos(u)}{2} \frac{du}{2}$$

$$= \frac{\theta}{2} \Big|_0^{\frac{\pi}{2}} + \frac{1}{4} \int_0^{\frac{\pi}{2}} \cos(u) du$$

$$= \frac{\theta}{2} \Big|_0^{\frac{\pi}{2}} + \frac{1}{4} \left[ \sin(u) \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left[ \frac{\pi}{2} - 0 \right] + \frac{1}{4} \left[ \sin(\pi) - \sin(0) \right]$$

(c)  $\int_0^{\frac{\pi}{4}} \sec^2 \theta \tan \theta d\theta$

$u = \tan \theta, du = \sec^2 \theta d\theta$   
 $u(\frac{\pi}{4}) = 1$   
 $u(0) = 0$

$$= \int_0^1 u du$$

$$= \frac{u^2}{2} \Big|_0^1$$

$$= \left[ \frac{1}{2} - \frac{0}{2} \right]$$

$$= \frac{1}{2}$$

$$= \frac{1}{2} \cdot \frac{\pi}{2} + \frac{1}{4} [0 - 0] = \frac{\pi}{4}$$